



SUJET STATAPP 314 - 2026-2027

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Organisme : Ekimetrics

Catégorie : Autre

Sujet :

Why Is the Model Uncertain? Explaining and Reducing Predictive Uncertainty in Machine Learning

Description du sujet :

Machine-learning models are generally designed to provide a prediction, for example, the expected selling price of a house. In many applications, however, the prediction alone is insufficient: it is also important to know how reliable it is.

Uncertainty-quantification methods address this problem by associating predictions with a range of plausible outcomes. For example, instead of predicting that a house will sell for ?400,000, a model may provide a prediction interval of ?360,000??450,000.

Such an interval tells us how uncertain the model is, but not necessarily why it is uncertain. For a particular observation, uncertainty may be high because important information is missing, because the observation is unusual compared with the training data, or because the outcome is intrinsically difficult to predict. Distinguishing these situations is important: if uncertainty results from missing information, collecting an additional variable may substantially improve the prediction.

The project asks:

> Can we identify which input variables are responsible for predictive uncertainty, and use these explanations to decide which additional information should be collected?

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The first objective will be to compare explanations of the prediction itself with explanations of its uncertainty. A variable that strongly affects the predicted value does not necessarily explain why the prediction interval is wide.

The second objective will be to test whether uncertainty explanations are actionable. Some variables will be deliberately hidden from the model at prediction time. An explanation method will recommend which missing variable to reveal. Students will then measure whether observing this variable actually reduces uncertainty while preserving valid predictive coverage.

This provides a quantitative test of uncertainty explanations: rather than asking only whether an explanation appears plausible, the project asks whether following it improves the information available for a prediction.

A controlled synthetic experiment will complement the real dataset. It will allow students to generate situations where some variables determine the expected outcome while others determine its variability, so that the true source of uncertainty is known.

Données du sujet :

The main application will use the Ames Housing dataset, which contains 2,930 residential property sales with more than 80 variables describing characteristics such as size, location, construction quality, basement and garage, along with the final sale price.

The richness of the covariates makes it possible to simulate different information states: some property characteristics can initially be hidden and subsequently revealed, mimicking the acquisition of additional information before making a final valuation.

A synthetic dataset will complement the real application and provide experiments in which the uncertainty-generating mechanism is known exactly.

Students may also propose another public regression dataset related to an application they are particularly interested in, provided that it supports a meaningful information-acquisition problem and is validated at the beginning of the project.

Modèles envisagés du sujet :

- Linear regression and tree-based regression models.
- Quantile regression and conformal prediction for predictive intervals.
- Standard feature-attribution methods such as SHAP.
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Bibliographie du sujet :

- De Cock, D. (2011). Ames, Iowa: Alternative to the Boston Housing Data as an End of Semester Regression Project. Journal of Statistics Education.
- Angelopoulos, A. N., & Bates, S. (2023). Conformal Prediction: A Gentle Introduction. Foundations and Trends in Machine Learning.
- Il Idrissi, M., Fernandes Machado, A., Charpentier, A., & Gallic, E. (2025). Unveil Sources of Uncertainty: Feature Contribution to Conformal Prediction Intervals.
- Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. NeurIPS.
- Molnar, C. Interpretable Machine Learning.

Notation du sujet :

Commentaires :